

Math 164 Midterm II Problem 1 Solution

Leo Liang

Note: When I say “point”, you can always substitute it with “vector” as every point can be treated as a vector. I just say “point” because it gives a little better intuition in understanding what is happening.

Problem 1

(a) Find the length of the curve C given by $r(t) = ti + 3\sin(2t)j + 3\cos(2t)k$ for $0 \leq t \leq 2\pi$

Goal: Find the arc length from a starting $t = 0$ to an ending $t = 2\pi$ with the given vector function $r(t)$.

Tools and Thought Process: We know that arc length is given by $L = \int_a^b |r'(t)| dt$. So we first find $r'(t) = \langle 1, 6\cos(2t), -6\sin(2t) \rangle$. Then $|r'(t)| = \sqrt{1^2 + 36\cos^2(2t) + 36\sin^2(2t)} = \sqrt{37}$. So the arc length from $t = 0$ to $t = 2\pi$ is $L = \int_0^{2\pi} \sqrt{37} dt = 2\pi\sqrt{37}$.

Interpretation: Starting at time $t = 0$ to $t = 2\pi$, as we travel along the curve given by $r(t)$ we have travelled a total distance of $2\pi\sqrt{37}$ units.

(b) Find the arc length function $s(t)$ for $r(t) = ti + 3\sin(2t)j + 3\cos(2t)k$.

Goal: We want a function that takes in a time t and outputs a scalar value that represents the arc length from some initial time $t = a$ to the input time t .

Tools and Thought Process: From our English statement in the Goal section, we can translate it mathematically to

$$s(t) = \int_a^t |r'(u)| du$$

where $t = a$ is a pre-set initial time and t is the input value of the function. Note that we write $r'(u)$ instead of $r'(t)$ in the equation (we simply just swap t with u) because we don't want to confuse the limits of integration with the vector function itself. Another way to think about it is that we want to find a function $s(t)$ from another known function $r'(t)$, so they should not share the same input parameter. Solving it out we have

$$s(t) = \int_a^t \sqrt{1^2 + 36\cos^2(2u) + 36\sin^2(2u)} du = \sqrt{37}t - \sqrt{37}a$$

Interpretation: We've constructed a function $s(t)$ that takes in an input t and outputs the distance travelled in terms of a , the starting time. Note that from the order of the limits of integration (t being on top and a on the bottom), we've assumed that $t > a$.

(c) Reparameterize $r(t) = ti + 3\sin(2t)j + 3\cos(2t)k$ in terms of arc length s measured from $t = 0$ in the direction of increasing t .

Goal: We want to construct a function $r(s)$ that takes in an input arc length s and outputs the position vector after traveling s units along the curve from the starting point given by $t = 0$. From just reading the problem, you should know that this is exactly what part (b) solved for, except now we are given the the initial time of $t = 0$. Note the importance of the phrasing in the problem “in the direction of increasing t ”. This tells us that the variable t is the upper limit of integration, just like how we have it in part (b). If the problem said “in the direction of decreasing t ”, we would have to recalculate part (b) but with t in the lower limit of integration.

Tools and Thought Process: Because the conditions of this problem are compatible with the result that we found in part (b), we simply let $a = 0$ and we have

$$s(t) = \sqrt{37}t - \sqrt{37} \cdot 0 = \sqrt{37}t.$$

We now have a relationship between $s(t)$ and t . So, we can express t in terms of $s(t)$ by rearranging our equality and we have $t = \frac{s(t)}{\sqrt{37}}$. We can substitute this back into our equation for $r(t)$ and obtain:

$$r\left(\frac{s(t)}{\sqrt{37}}\right) = \frac{s(t)}{\sqrt{37}}i + 3\sin\left(\frac{2s(t)}{\sqrt{37}}\right)j + 3\cos\left(\frac{2s(t)}{\sqrt{37}}\right)k.$$

So far, we've done exact algebraic rearrangement and substitution, something you should be comfortable with. But, as mathematicians, we should immediately recognize that there are redundant expression in here that should be removed. Here were my observations:

- Firstly, the expression $r\left(\frac{s(t)}{\sqrt{37}}\right)$ makes no sense because a function should take in an input, not an expression. Remember that our goal was to construct a function that takes in an input arc length s .
- Secondly, from the context of the problem, the expression $s(t)$ makes no sense because the arc length will be given by the input s ; in other words, s is NOT dependent on t . Or you could also say, s is NOT a function of t in this context.

From these observations, we clean up the equation found before and now have

$$r(s) = \frac{s}{\sqrt{37}}i + 3\sin\left(\frac{2s}{\sqrt{37}}\right)j + 3\cos\left(\frac{2s}{\sqrt{37}}\right)k.$$

This is the the only answer that will be accepted on the test, as to write it the first way $r\left(\frac{s(t)}{\sqrt{37}}\right) = \frac{s(t)}{\sqrt{37}}i + 3\sin\left(\frac{2s(t)}{\sqrt{37}}\right)j + 3\cos\left(\frac{2s(t)}{\sqrt{37}}\right)k$ completely takes away from the meaning of the problem and does not demonstrate your mathematical understanding.

Interpretation: We've re-parameterized our initial function $r(t)$ to the function $r(s)$ which takes in an input arc length s and outputs the position vector after we've travelled the distance s from the starting point given by the time $t = 0$.

(d) Starting at $t = 0$, at which point on the curve $r(t) = ti + 3\sin(2t)j + 3\cos(2t)k$ are we after traveling a distance of $\frac{\sqrt{37}\pi}{6}$.

Goal: We want to find where we are on the curve $r(t)$ after traveling a distance of $\frac{\sqrt{37}\pi}{6}$ from the point corresponding to $t = 0$. Note that our answer should be a vector, namely the position vector.

Tools and Thought Process: Some may be tempted to immediately apply the result from part (c), but you should always justify to yourself why it is appropriate. Always check your assumptions. Here is the justification:

- The curve $r(t)$ is the same vector function used to derive the equation for part (c). If the vector function was not the same, you would have to re-derive the $r(s)$ function.
- We are starting at $t = 0$, which is the assumption of the equation we derived in part (c). If we are starting at $t = t_0 \neq 0$, you would have to re-derive the $r(s)$ function. You would use the equation derived in part (b) but with $a = t_0$.
- The problem should've explicitly stated "*in the direction of increasing t* " but we can assume that is the what the professors meant. I would ask on the test. If the problem said "*in the direction of decreasing t* ", you would have to redo the integral in part (b) but the limits of integration would be from t to a .

Since the conditions of this problem are compatible with the result that we found in part (c), we can simply input the distance $s = \frac{\sqrt{37}\pi}{6}$ and we have

$$r\left(\frac{\sqrt{37}\pi}{6}\right) = \frac{\pi}{6}i + 3\sin\left(\frac{\pi}{3}\right)j + 3\cos\left(\frac{\pi}{3}\right)k = \frac{\pi}{6}i + \frac{3\sqrt{3}}{2}j + \frac{3}{2}k.$$

Interpretation: Starting at $t = 0$, after traveling a distance of $\frac{\sqrt{37}\pi}{6}$ units on the curve $r(t)$ in the direction of increasing t , we are at the point $\left(\frac{\pi}{6}, \frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$.